Mathematics - K-12 Level Assessment 14 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. A meteorologist reports that there is a 30% probability of rain and no sun. If there is a 40% probability of no rain, then the probability of both rain and sun is
   (a) 0.16
   (b) 0.24
   (c) 0.3
   (d) 0.5

2. When $\sqrt[3]{-128}$ is simplified, the result is $a\sqrt[3]{b}$, where a is an integer, and b is a positive integer. If b is as small as possible, then what is $a + b$?
   (a) 2
   (b) -4
   (c) 6
   (d) -2

3. Simplify 5^3. Use a calculator, paper and pencil, or mental math.
   (a) 15
   (b) 125
   (c) 8
   (d) 53

4. Find the quotient of 5.04/0.7
   (a) 0.72
   (b) 0.702
   (c) 10.0571
   (d) 7.2

5. Marguerite earned a score between 75 and 89 on all of her previous spelling tests. She earned a score of 100 on her next test. Which of the following statements is true?
   (a) The mode will increase.
   (b) The mean will increase.
   (c) The mean will decrease.
   (d) The median will decrease.

True / False (10 marks)

Answer True or False

6. True or False: The probability of picking a blue chip from the bag is 5 out of 15.
7. True or False: There are 999 three-digit positive integers ranging from 100 to 999.
8. True or False: The number of heads in three coin tosses can be modeled using a binomial distribution.
9. True or False: 0.32 grams is equal to 3.2 centigrams.
10. True or False: The ratio 24 over 64 forms a proportion with 3 over 8.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Rob feeds his cats using 1 box of cat food every 5 days. Calculate the number of boxes he would need in a month, and explain your reasoning and calculations.
12. Discuss the scenario of randomly selecting three points on a circle. Analyze the conditions under which all pairwise distances between these points are less than the circle's radius. What is the probability of this occurrence?

End of Paper